Exceptional times for nearest-neighbor exclusion percolation on the triangular lattice have Hausdorff dimension 31/36

Abstract

We prove the 31/36 exceptional-time theorem for nearest-neighbor exclusion percolation on the triangular lattice. The proof compares two-time crossing probabilities for conservative dimer updates and independent refresh, using a local reference response to normalize time. A forest estimate controls every degree of the expansion; a sewing argument identifies each fixed degree. The resulting annular mixing estimate gives the energy bound needed for exceptional times. A Lean formalization accompanies the proof.

1 The question and the statement

Identify the triangular lattice with $\mathbb{T} = \{m + ne^{i\pi/3} : m, n \in \mathbb{Z}\}$, with edges between points at distance one. Initially, sites are independently black or white with probability 1/2. Each unordered edge has an independent rate-one Poisson clock; when it rings, its two endpoint colors are exchanged. Write ω_t for the configuration and

$$\mathcal{E} = \{t \geq 0 : \omega_t \text{ has an infinite black cluster}\}.$$

At every deterministic time the configuration is critical Bernoulli percolation, so it has no infinite cluster almost surely. The question concerns the random set $\mathcal{E}$.

Broman, Garban and Steif asked whether conservative percolation has exceptional times [1, Section 9.4]. Garban and Vanneuville later formulated the dimension-31/36 conjecture for symmetric exclusion dynamics [2, Conjecture 1.7]; it appears in their May 2016 preprint. Nearest-neighbor exchanges are a principal case of that conjecture. For independent dynamical percolation the dimension is 31/36 [3, Theorem 1.4(1)], while the exclusion upper bound is already known [2, Proposition 3.3]. We prove the nearest-neighbor case on $\mathbb{T}$.

Theorem 1. *Almost surely, simultaneously for all* $0 \leq a < b < \infty$,

$$\dim_{\mathrm{H}}(\mathcal{E} \cap (a, b)) = \frac{31}{36}.$$

In particular, every such interval contains exceptional times.

2 Proof

1. A local clock and a comparison of finite crossing words

Put $\varepsilon = 2^{-n}$ and $a_\varepsilon(r) = \mathbb{P}(A_4(r, 1))$ on the *same* mesh ε. Here $A_4(r, R)$ denotes four disjoint arms of alternating colors between the circular cuts at radii r and R, using closed-cell paths. We use RSW, arm separation and quasi-multiplicativity in the forms of [4, Sections 2.3 and 4, Propositions 15–16], with the fixed-shape comparisons of [9, Section 2].

Fix a sufficiently large microscopic constant B. Separate four alternating arms at the microscopic cut and at a fixed positive radius. Disjoint RSW corridors attach their outer ends to the left and right black sides and the upper and lower white sides of $Q_0 = [-1, 1] \times [-1/2, 1/2]$. A fixed finite collar routes the inner ends around the neighboring pair, leaving its two bits free; finite energy

pays only a constant factor for its prescribed colors. Jordan separation [13, Section 4.1] excludes bypasses across the white connections, so the crossing reads the first free bit for all four assignments of the pair. Arm separation and quasi-multiplicativity therefore give probability at least $c\,a_\varepsilon(B\varepsilon)$, uniformly over a fixed bulk range. This is an ordinary one-time attachment estimate.

We also use the ordinary two-source sewing construction of [9, Section 6, Lemmas 6.5–6.6 and the relative many-trial estimate]. For a fixed finite polygon family and a compact, positively separated insertion range, it supplies annular success events invariant under source exchange and the relevant inside exchange. Success makes the crossings depend on the enclosed colors through a single join bit. For two independent ordinary fair fields ω, η, the first-point failure event N satisfies

$$\mathbb{P}(N \cap A_4^{\mathrm{mic}}(\omega) \cap A_4^{\mathrm{sq}}(\eta)) \le \epsilon\, a_\varepsilon(B\varepsilon) a_\varepsilon(h). \tag{1}$$

The two arm events refer respectively to the microscopic insertion and the fixed square. To apply the source estimate, trim their paths to buffered graph cuts inside the protected annulus; the resulting arms imply the source's probe event. Fixed-shape arm comparisons change its normalization only by a constant factor. The relative many-trial bound has the form $\rho + C/n$: choose ρ small, fix the microscopic enlargement, then choose the finite trial count n large. Next fix h below those trials and finally take a sufficiently fine mesh. This gives (1), uniformly on the compact separated location range. The reference-square influence is at least $c\,a_\varepsilon(h)$, by the specialization of [9, Lemma 6.1], again using the fixed square/circle comparison. These estimates apply on the triangular lattice as stated in [9, Sections 2 and 6].

Use integer coordinates on $\mathbb{T}$. For each phase $\sigma \in \{0,1\}$, pair $(2m+\sigma, n)$ with $(2m+\sigma+1, n)$. Consider two rate-one updates on each pair, independently between pairs: C replaces its colors uniformly subject to their black count; M chooses an endpoint uniformly and refreshes that bit fairly. Thus C keeps or swaps the pair with equal probability, whereas M is independent site refresh at rate $1/2$.

For $A \in \{C, M\}$, let $S_A(\varepsilon)$ be the expected change indicator of the Q_0 crossing when one A update is applied to the central horizontal pair, with all other colors an ordinary fair field. Define

$$J_A(\varepsilon) = \frac{S_A(\varepsilon)}{2\varepsilon^2}, \qquad u_A(\varepsilon) = \frac{\varepsilon^2}{S_A(\varepsilon)}. \tag{2}$$

The attachment estimate and the exact two-bit calculation give $S_A \ge c\,a_\varepsilon(B\varepsilon) > 0$. The four-arm exponent $5/4$ [6, Theorem 4] consequently gives, for every $0 < \eta < 3/4$ and all sufficiently small meshes,

$$u_A(\varepsilon) \le C_\eta \varepsilon^{3/4-\eta}. \tag{3}$$

Translation of the reference rectangle identifies the response at every insertion with this same S_A.

Let F_ε be a Boolean word in the literal closed-cell crossings of a fixed finite family of dyadic polygonal quads. Here the quads are images of dyadic axis-parallel grid quads under $(u,v) \mapsto u + e^{i\pi/3}v$; invariance of domain [15, Theorem 2B.3] identifies their polygonal interiors. Put $q_\varepsilon^A(t) = \mathbb{P}(F_\varepsilon(\omega_0) = F_\varepsilon(\omega_t) = 1)$ for the corresponding matching dynamics.

Lemma 2. *For each fixed $T > 0$ and each phase,*

$$q_\varepsilon^C(T u_C(\varepsilon)) - q_\varepsilon^M(T u_M(\varepsilon)) \longrightarrow 0. \tag{4}$$

The second term is the independent site-refresh probability at time $T\varepsilon^2/(2S_M(\varepsilon))$.

Proof. Take a fixed physical box containing every tested quad and cover its sites by the matching. The two-time law of one pair has an exact expansion

$$p_A(t) = p_0 + t\{w_A + r_A(t)\}, \qquad \sum w_A = \sum r_A(t) = 0, \quad \|r_A(t)\|_{\mathrm{TV}} \le t \quad (0 < t \le 1).$$

Here $\|v\|_{\mathrm{TV}} = \sum_h |v(h)|$ is the full variation norm, and p_0 is the fair static pair, constant between the two times; w_A is the derivative of the actual pair law at zero. Direct calculation gives $\|w_C\|_{\mathrm{TV}} = 1/2$ and $\|w_M\|_{\mathrm{TV}} = 1$.

For an unordered selection S of pairs, integrate arbitrary zero-mass signed weights w_x at S and the static law elsewhere. The absolute value of the resulting kernel is bounded by

$$G_\varepsilon(S) \prod_{x \in S} \|w_x\|_{\mathrm{TV}}, \qquad G_\varepsilon(S) \geq 0. \tag{5}$$

Centering forces a local change at every selected pair. Away from the quad boundaries this forces four alternating arms; near a side it forces the appropriate half-plane arms [9, Lemma 2.2]. Record successive mergers of the probe neighborhoods as a rooted forest. RSW, arm separation, the four-arm power bounds and the half-plane three-arm bound [4, Theorems 20 and 23(ii)] control the disjoint probe costs.

The two quantitative consequences are as follows. Choose a lower four-arm power $g \in (5/4, 2)$, for example $g = 3/2$, and set $\beta = 1 - g/2 > 0$. For some $A < \infty$, uniformly in phase and update rule, eventually *for every* $k \geq 0$,

$$u_A(\varepsilon)^k \sum_{|S|=k} G_\varepsilon(S) \leq A^k k^{-\beta k}, \tag{6}$$

where the right side is one at $k = 0$. The forest integration uses the stopping radius $R_k = \min\{R_*, c_*/\sqrt{k}\}$. Packing ensures that the microscopic cutoff lies below R_k for every nonempty selection; larger degrees have no selections. Counting the merge forests and integrating their radial edges gives (6), including the selection factorial. Moreover, for fixed k the normalized ordered kernels have a uniform L^p forest bound for any $p > 1$ sufficiently close to one that $gp < 2$.

Write $K_A(\varepsilon; \boldsymbol{x})$ for the kernel with k raw insertions w_A. On tuples separated from each other and from the polygon frontiers, the sewing estimate gives

$$\sup_{\boldsymbol{x}} \left| \frac{K_C(\varepsilon; \boldsymbol{x})}{S_C(\varepsilon)^k} - \frac{K_M(\varepsilon; \boldsymbol{x})}{S_M(\varepsilon)^k} \right| \longrightarrow 0. \tag{7}$$

Indeed, hold the other $k-1$ signed insertions fixed and compare the distinguished insertion and the translated reference crossing with one common forced square. Put $a = a_\varepsilon(B\varepsilon)$ and $b = a_\varepsilon(h)$. If H_h is the square kernel and D_h its reference response, write $\widehat{K}_A$ for the kernel with distinguished rule A and the other rules unchanged. The two errors are $E_A = \widehat{K}_A D_h - H_h S_A$. The source symmetries and disjoint ordinary probes give $|E_A| \leq C_k \epsilon a^k b$. The exact cancellation

$$D_h(\widehat{K}_C S_M - \widehat{K}_M S_C) = S_M E_C - S_C E_M$$

and the bounds $D_h \geq cb$, $ca \leq S_A \leq Ca$ compare the two normalized insertions. The extra $k-1$ probe cutoffs are fixed independently of the tolerance. Telescoping replaces all k entries. For each tolerance one chooses the finite trials, then a sufficiently small fixed h, and finally sends ε to zero. The normalized microscopic difference is independent of h, so this proves (7).

The degree-k coefficient after clock normalization is exactly

$$c_{A,k}(\varepsilon) = \frac{\varepsilon^{2k}}{k!} \sum_{\boldsymbol{x} \text{ distinct}} \frac{K_A(\varepsilon; \boldsymbol{x})}{S_A(\varepsilon)^k}. \tag{8}$$

Remove tuples within distance δ of a collision or a polygon frontier. Their scaled counting mass tends to zero with δ; the L^p forest bound and Hölder's inequality make their contribution uniformly

negligible. On the remaining tuples use (7). Thus $c_{C,k} - c_{M,k} \to 0$ for each fixed k, including the common static degree zero.

Finally, (5) also applies to the centered remainders r_A. By (3), replacing raw insertions by the effective insertions changes each fixed-degree coefficient by $o(1)$. Equation (6) bounds the degree-k terms at time T by $(2AT)^k k^{-\beta k}$, a summable sequence. Dominated summation therefore proves (4) for the actual finite product laws, including all repeated-update remainders. The identification of M with site refresh at half its pair time is exact. □

2. From crossing words to an energy bound

The dense quad coordinates and arm-event convergence of [7, Definition 2.1, Lemma 2.9 and Corollary 2.10] approximate a fixed hexagonal four-arm event in static probability by a finite crossing word: first approximate its limiting indicator by a finite cylinder, then pass to sufficiently fine meshes. The word is fixed before the mesh threshold. Stationarity bounds the error in a two-time probability by twice the static error. Apply Lemma 2, followed by full independent mixing [8, Theorems 1.3(i), 1.4(i) and Remark 1.5, $\gamma = 0$]. The local independent clock is admissible because $S_M(\varepsilon) \le C\alpha_4(\varepsilon, 1)$ by ordinary pivotal necessity; hence its site time is at least a fixed multiple of $T\varepsilon^2/\alpha_4(\varepsilon, 1)$. Same-event independent correlations decrease with time by their Fourier expansion, placing the local clock in the mixing regime.

The square of the static four-arm probability of an annulus of aspect M is at most $CM^{-9/4}$. Choose a sufficiently large dyadic M, then a sufficiently large fixed T, then a sufficiently fine mesh. The resulting conservative four-arm bound is at most $(2M)^{-17/8}$. Its reference time at lattice scale s is bounded by $C_\eta s^{-(3/4-\eta)}$, by (3). Buffered disjoint annuli and monotonicity in time iterate this fixed-aspect estimate. Dyadic rounding produces a decorrelation length

$$m(t) \le C_\eta t^{-1/(3/4-\eta)}, \qquad 0 < t < 1, \tag{9}$$

above which the paired four-arm probes decay with exponent $17/8 > 2$.

Let $A_R = \{0 \leftrightarrow \partial B_R\}$ and $\alpha_1(R) = \mathbb{P}(A_R)$. For canonical matching dynamics write $q_R^C(t) = \mathbb{P}(A_R(\omega_0) \cap A_R(\omega_t))$. For a wired annular connection from r to R, average the exterior replicas independently and retain canonical dependence only in a buffered middle annulus. Write q_{mix} for this mixed two-time probability and $a(r, R)$ for its ordinary one-time mean. Finite-cluster boundary separation is used in the form of [14, Section 5.5]. The exact conditional independent-replica representation permits the finite Efron–Stein inequality [10, Proposition 1]. A changed conditional connection probability forces a paired four-arm probe. All active cells lie in the buffered bulk, and the exponent $17/8 = 2 + 1/8$ gives

$$q_{\mathrm{mix}} - a(r, R)^2 \le C\big(m(t)/r\big)^{1/8} q_{\mathrm{mix}}.$$

The duplicated-event Harris inequality and fixed black collars give $q^C(W(r, R)) \le c^{-8} q_{\mathrm{mix}}$ for the canonical wired event. Choose N so that the variance coefficient is at most $1/2$ when $r \ge Nm(t)$. Absorption at $r = 2Nm(t)$ bounds this canonical probability by $2c^{-8}a(r, R)^2$. The disjoint inner and outer matching labels and ordinary one-arm quasi-multiplicativity then yield

$$q_R^C(t) \le C\,\frac{\alpha_1(R)^2}{\alpha_1(Nm(t))}$$

in the large-radius case; the static bound handles smaller radii. The one-arm exponent and quasi-multiplicativity [5, 4], together with (9), give, for every $\zeta > 0$,

$$q_R^C(t) \le \min\{\alpha_1(R),\, C_\zeta \alpha_1(R)^2 t^{-5/36-\zeta}\}. \tag{10}$$

The numerical exponent is $(5/48)/(3/4) = 5/36$. All finite covers and both matching phases are included in these estimates.

3. Nearest-neighbor exclusion and exceptional times

Let H_C be the positive generator of canonical matching dynamics and H_X that of rate-one nearest-neighbor swaps, on a finite edge truncation containing the event's matching-closed support. On a matched edge,

$$I - \Pi = \tfrac{1}{2}(I - \mathsf{Swap}), \qquad H_X \geq H_C \geq 0$$

as quadratic forms. For $0 < \gamma < 31/36$, choose $\zeta > 0$ with $\gamma + 5/36 + \zeta < 1$. Equation (10) gives

$$\sup_R \alpha_1(R)^{-2} \int_0^1 t^{-\gamma} q_R^C(t)\, \mathrm{d}t < \infty.$$

Fractional resolvent order transfers this bound to exclusion. More explicitly, $(I + H_X)^{\gamma-1} \leq (I + H_C)^{\gamma-1}$; the Laplace representation, followed by monotonicity of the canonical correlation for the tail $t \geq 1$, gives

$$\int_0^1 t^{-\gamma} q_R^X(t)\, \mathrm{d}t \leq (e+1) \int_0^1 t^{-\gamma} q_R^C(t)\, \mathrm{d}t \leq C_\gamma \alpha_1(R)^2. \tag{11}$$

Backward-trace stabilization, finite-cylinder convergence and Fatou's lemma pass the bound along an edge exhaustion to the actual infinite process; the containment threshold may depend on R.

On $I = [0,1]$, consider the normalized occupation measures

$$\nu_R(\,\mathrm{d}t) = \alpha_1(R)^{-1} \mathbf{1}_{A_R(\omega_t)} \mathbf{1}_I(t)\, \mathrm{d}t.$$

Their expected mass is one. Stationarity and (11) bound their expected γ-energy uniformly in R; energy also bounds squared mass on I. A second-moment and Markov argument therefore gives, with positive probability, a common positive mass lower bound and finite energy upper bound for infinitely many radii. Apply the compact-set capacity criterion [11, Lemma 6.2] to the decreasing closures of the arm-time sets. Their intersection has dimension at least γ on this measurable witness event.

Closing each site's black-time set gives a configuration with an infinite cluster at every time in this compact intersection. There are almost surely no simultaneous rings of distinct edge clocks. Local path regularity shows that this configuration differs from the actual one at at most the two endpoints of the edge ringing at that time. Finite recoloring preserves existence of some infinite cluster on a locally finite graph. Thus the compact intersection consists of actual exceptional times, although its clusters need not contain the origin.

The product construction from Poisson marks [12] and graphical regularity [2, Appendix A] is equivariant under lattice translation and has an ergodic unit spatial shift. Saturating the measurable positive-probability witness event under this shift upgrades the lower bound to probability one. Affine pullback to a time window $a + c[0,1]$, $0 < c \leq 1$, changes the energy bound by at most $c^{\gamma-1}$. Intersect over rational $\gamma \in (0, 31/36)$ and rational windows; inclusion of rational subwindows gives the simultaneous lower bound for every real $0 \leq a < b$. The upper bound follows from [2, Proposition 3.3], after the proved constant time change from its normalized rate convention. Hausdorff dimension is invariant under that positive linear time change. This proves Theorem 1.

Formalization. The accompanying Lean 4.34.1 development verifies the argument from 24 explicit formulations of the cited results, together with positivity of the one-arm decay exponent. The literature results and their specializations explained above are supplied as hypotheses; the new comparison, coefficient limits, forest bounds, clock estimates and exceptional-time argument are proved within Lean. The entry point is `exists_realization_dimension` in the module `PublishedNearestNeighborDimension`. The complete build checks 443 source files, with only `propext`, `Classical.choice` and `Quot.sound` as transitive axioms. The accompanying archive `nearest-neighbor-31-36-lean.zip` contains the precise input formulations, source correspondences and reproducible build record.

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